

Finger-to-Palm: Single-Handed Touch Gestures Using the Palm

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Figure 1: Using Finger-to-Palm touch commands, the user adjusts the volume of a movie in a mixed-reality headset by pressing the ring and pinky fingers onto the palm.

Abstract

This paper introduces the Finger-to-Palm touch interaction, a new interface for enhancing head-mounted display interactions. It bypasses the precise targeting required by thumb-to-finger methods by directly mapping touch points on the palm to HMD functions. A sticker device with pressure sensors enables an unobtrusive setup. An initial study evaluated various gestures for their physical and mental demands and user confidence, selecting 10 gestures. These were rigorously tested in a follow-up study using our prototype, focusing on completion time, error rates, and user preferences. This led to the recommendation of seven gestures—four fingers, index-middle-ring, index-middle, single index finger, single middle finger, middle-ring-pinky, and middle-ring—chosen for their effectiveness and user acceptance, providing a foundation for developing efficient finger-to-palm touch commands in future applications.

CCS Concepts

• Human-centered computing → Gestural input.

Keywords

Multi-touch, gesture input

ACM Reference Format:

Ming-Yun Hsu and Liwei Chan. 2025. Finger-to-Palm: Single-Handed Touch Gestures Using the Palm. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720213>

1 Introduction

In Human-Computer Interaction, there is increasing interest in on-body or skin-based touch interfaces for Head-Mounted Displays (HMDs). These interfaces, which allow quick access to features without visual attention by utilizing human spatial awareness and muscle memory, are being explored on various body parts like the arm, lap, and face [3, 9, 15–18, 25, 34, 36, 37, 39]. They focus particularly on interfaces operated solely by finger motions, eliminating the need for broader hand movements. A notable format is thumb-to-finger interaction, where different finger segments perform shortcut functions similar to a keyboard. However, implementing this involves challenges, such as using wearable sensors like finger rings or wristbands to ensure effective coverage across multiple finger areas.

In this work, we introduce the "Finger-to-Palm interface," a novel on-body touch interface designed as a sticker device for the user's palm. It leverages the palm's natural contact with the fingers during fist-making to enable a range of multi-touch gestures, activating shortcut functions in Head-Mounted Displays (HMDs). For instance, users can adjust HMD functions such as volume by touching the ring and pinky fingers to the palm without needing traditional controllers or menus. This design offers a potential advantage over the thumb-to-finger input method. It simplifies interaction by removing the need for precise thumb targeting, which some users find

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720213>

challenging. Instead, each finger's natural touch point on the palm directly corresponds to a specific action, streamlining operation.

Our research involved two studies to evaluate and refine this interface. The first study assessed various gestures for their physical and mental demands and user confidence, leading to the selection of 10 optimal gestures. The second study further tested these gestures using a prototype device, ultimately recommending seven specific gestures: four fingers, index-middle-ring, middle-ring-pinky, index-middle, middle-ring, index alone, and middle alone. These selected gestures form the basis for developing effective finger-to-palm touch commands.

2 Related Work

Skin and on-body touch interfaces significantly enhance human-computer interaction by making it more intuitive, accessible, and seamless, turning the human body into an interactive platform. These technologies not only boost mobility but also improve multitasking abilities. Our review, consistent with our study's scope, highlights a range of body locations and applications within skin interface research, as detailed in related works [2, 14, 30]. Research has shown the efficacy of using human body surfaces for interactive technologies. Initiatives like PUB [15], TapSkin [36], and ActiTouch [37] have utilized the arm for touch inputs, while projects like Lap-Touch [17] have explored the lap as an interactive zone. The face and ear have also been examined as unique touch interaction sites in studies such as HandToFace [25], FaceSight [34], FaceSense [9], and EarTouch [10], demonstrating the versatility of potential touch interfaces across the body.

The development of on-body interfaces increasingly focuses on utilizing the skin of the hand as an interactive surface [6, 20–22, 27, 38, 40]. Among these, finger-based interfaces are particularly compelling due to their ability to enable one-handed interaction [26], allowing users to quickly access interactive functionalities [7]. A key gesture in this domain is the pinch between the thumb and index finger, which is essential for midair virtual reality (VR) interactions and widely adopted in consumer HMDs equipped with gaze tracking [19, 23, 24]. Recent advancements have further refined these interactions, introducing enriched thumb-to-index gestures [5, 28, 31] and pinch-based text entry supported by predictive text [8]. Additionally, the palm provides a larger surface for touch interfaces than fingers, useful for controlling devices remotely [4, 13, 35] and offers a less crowded keyboard layout when integrating finger and palm interactions [32]. However, palm-based interfaces usually require the use of the opposite hand [33]. The dorsal hand has been explored for touch inputs, though it also necessitates using the other hand [1, 11].

In this work, we explore the use of the palm as a site for single-handed touch inputs through finger-to-palm gestures. This approach involves using specific areas of the palm that detect touches from fingers in a fist position, allowing for quick and accessible command execution with one hand. Originally introduced by Nit-tala et al. in 'Multi-touch skin' [20], we build on this concept by offering practical guidelines for designing these gestures, informed by insights from our user studies.

3 System Implementation

The finger-to-palm touch recognition device features four pressure sensors (Interlink Electronics' FSR™ 400) encased between two layers of kinesiology tape, as shown in Figure 2. Positioned on the palm where the fingers naturally contact, the sensors are tailored to each user's unique finger positions. These sensors connect to a WIFI Arduino unit (Adafruit M0 WiFi module) that communicates wirelessly with a computer or through a serial port. The connected computer runs custom Unity software that interprets the gestures by analyzing pressure data from the sensors at 60 Hz.

The software detects user actions by analyzing changes in sensor pressure, triggering corresponding events based on these variations. It first establishes a baseline pressure for each sensor without touch. A touch gesture is only recognized as an event if sustained for over 150 milliseconds, reducing accidental activations. To avoid repetitive events from prolonged gestures, the software reports a new event only after the prior gesture is released.

4 STUDY 1: SELECTION OF TOUCH GESTURES

This study is designed to evaluate the viability of finger-to-palm gestures by assessing the participants' perceptions of physical and mental exertion, as well as their confidence in performing these gestures.

4.1 Apparatus

This experimental setup, as displayed in Figure 3, included a table, with a laptop placed on it administering study tasks through a study program written in Unity3D. Adjacent to the laptop, a tablet was provided for participants to enter responses to questionnaires associated with the task. Participants were seated at the table in a chair equipped with armrests, where they could comfortably place their dominant hand on either the armrest or their thigh. After participants assumed the hand position, the camera was set to focus on the hand placement, enabling the recording of hand movement videos throughout the study (Figure3b).

4.2 Procedure and Task

The study started with a briefing on the concept and tasks, followed by participants providing informed consent while seated. They used their dominant hand to perform finger-to-palm gestures, resting the hand on an armrest or thigh, palm down, as found preferable in a pilot study. Participants were shown abstract icons of the gestures (index, middle, ring, and pinky fingers) with the active finger highlighted in orange, and instructed to apply consistent pressure to achieve a distinct sensation. They completed the tasks at their own pace without time constraints. After finishing the tasks presented in random order, participants rated the physical and mental demands and their confidence in performing the gestures on a 7-point Likert scale, followed by a semi-structured interview to discuss responses and gather feedback. Each participant spent about an hour on the study, with two minutes between tasks.

4.3 Results and Discussion

We recruited 15 participants (4 females and 11 males) with a mean age of 22.5 years (SD: 1.59) via snowball sampling from a local university. Among the 15 participants, two were left-handed and

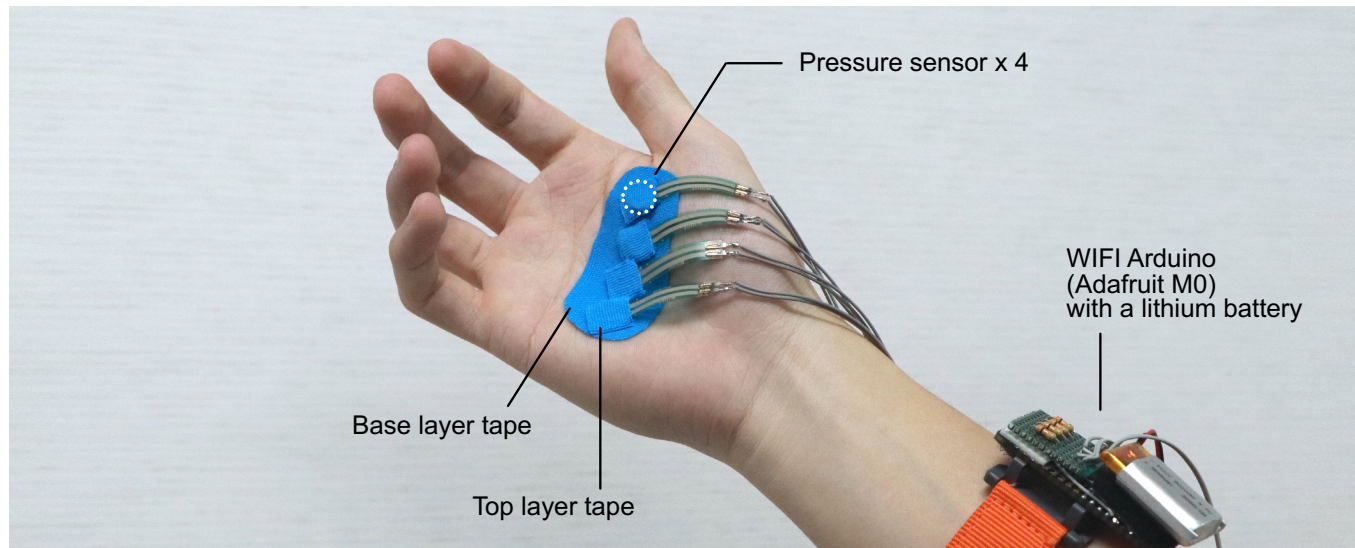


Figure 2: The Finger-To-Palm prototype consists of four pressure sensors strategically positioned on the palm, using base and top layers of kinesiology tapes.



Figure 3: Study 1 Setup: Participants were seated at a desk with their hands resting either on the armrests or in their laps. A laptop presented tasks, and a tablet positioned to the right was used to answer questions.

the rest were right-handed. Figure 5 shows user ratings for physical and mental demands, and confidence for the 15 gestures studied, using abstract notation for concise gesture labeling. The textual and iconic representations of these gestures are detailed in Figure 4.

Based on average ratings, the gestures with the lowest physical demand were middle-ring-pinky (M: 1.66, SD: 1.04), followed by full (M: 1.73, SD: 1.22), index (M: 1.86, SD: 1.18), middle-ring (M: 2.13, SD: 1.59), and middle (M: 2.26, SD: 1.53). The most physically demanding gestures were index-middle-pinky (M: 6.47, SD: 0.83), middle-pinky (M: 5.73, SD: 1.33), index-pinky (M: 4.67, SD: 1.71), index-ring-pinky (M: 4.27, SD: 1.39), and index-ring (M: 3.87, SD: 1.55). Gestures with higher demand often involved non-sequential finger use, creating ergonomic challenges. Conversely, gestures with lower demands typically featured sequential finger usage across different numbers of fingers, aligning better with ergonomic principles and reducing physical strain, as reported by participants.

In terms of mental demand, the gestures with the lowest were index (M: 1.4, SD: 0.82), full (M: 1.6, SD: 1.29), middle (M: 1.73, SD:

0.79), middle-ring-pinky (M: 1.87, SD: 1.06), and middle-ring (M: 2.06, SD: 1.09). The most mentally demanding were index-middle-pinky (M: 5.53, SD: 1.64), middle-pinky (M: 5.33, SD: 1.44), index-ring-pinky (M: 4.2, SD: 1.47), index-ring (M: 3.8, SD: 1.52), and index-pinky (M: 3.73, SD: 1.75), all featuring non-sequential finger use and rarely encountered in daily activities, increasing cognitive effort.

For confidence, the highest ratings were for index (M: 6.87, SD: 0.35), full (M: 6.6, SD: 1.05), middle-ring-pinky (M: 6.53, SD: 1.06), middle (M: 6.53, SD: 0.63), and middle-ring (M: 6.47, SD: 1.13). The lowest were index-middle-pinky (M: 2.53, SD: 1.77), middle-pinky (M: 3.53, SD: 1.68), index-pinky (M: 4.2, SD: 1.82), index-ring-pinky (M: 4.53, SD: 1.77), and index-ring (M: 4.87, SD: 1.59), consistent with those rated high in physical and mental demands, reflecting their complexity and infrequency in daily use.

The Friedman test revealed significant main effects for mental and physical demands, as well as confidence levels, across 15 gestures. However, post-hoc comparisons showed limited differences due to the large number of conditions. To refine the analysis, gestures were divided into two groups: the top ten gestures ranked by physical, mental, and confidence criteria (sequential-finger group) and the bottom five gestures (non-sequential-finger group). Wilcoxon signed-rank tests showed that sequential-finger gestures had higher confidence ($Z = 3.41, p < 0.001$), lower physical demand ($Z = -3.41, p < 0.001$), and reduced mental strain ($Z = -3.30, p < 0.001$) compared to non-sequential gestures.

In summary, sequential-finger gestures were ergonomically simpler, requiring less effort and boosting confidence, while non-sequential gestures were more demanding physically and mentally, with lower confidence levels. Based on these findings and participant feedback, we selected 10 gestures for further evaluation in Study 2, focusing on performance metrics.

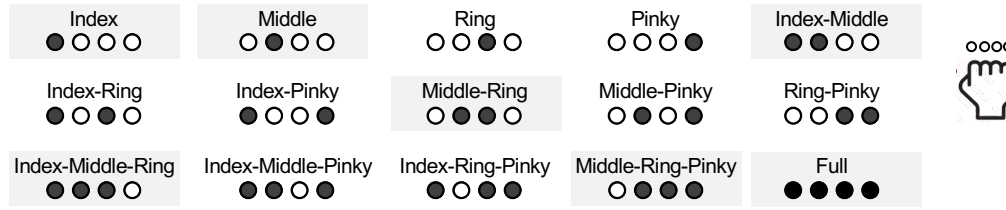


Figure 4: The iconic and textual labels for the fifteen gestures. Seven gestures recommended by our study result are highlighted in gray.

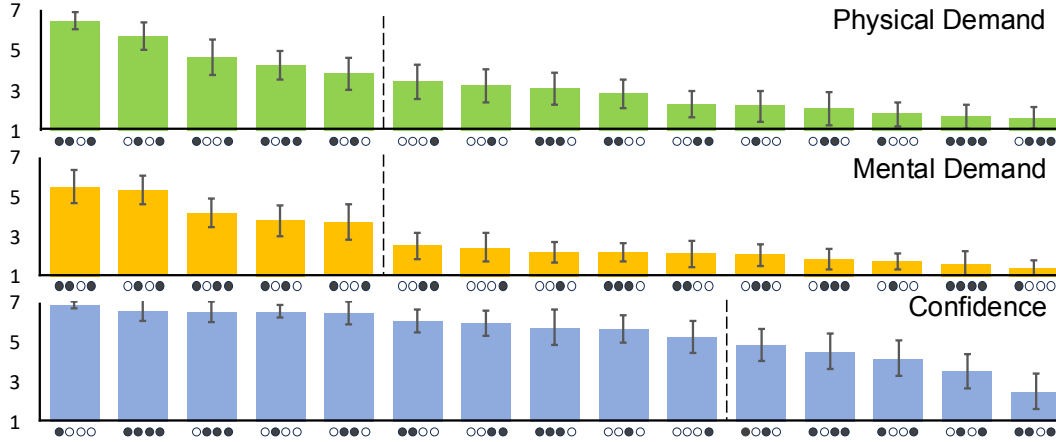


Figure 5: User ratings of physical demand, mental demand, and confidence for the 15 gestures. The error bars denote 1 standard error. Only the significant differences highlighted in the discussion of the result are indicated in the chart.

5 STUDY 2: USER PERFORMANCE

Building on Study 1's findings, ten gestures were identified as physically and mentally manageable for participants. Study 2 further evaluates these gestures using objective metrics like completion time and error rate, along with subjective assessments of user preference. The same setup from Study 1 was used.

5.1 Procedure and Task

After a briefing, participants were seated and assisted in wearing the finger-to-palm device, customizing sensor layouts to their unique touch positions (Figure 6). This involved applying a base layer of tape to the palm, marking touch positions with ink-dabbed fingers, and securing sensors with another tape layer. Participants chose comfortable hand positions and performed ten gestures displayed on a laptop in a counterbalanced order to minimize order effects. They practiced the gestures before the experiment, receiving immediate feedback from the sensors, which was disabled during the actual tests. After completing the tasks, participants selected their top five gestures and discussed their preferences in a semi-structured interview, with each session lasting about 50 minutes.

5.2 Results and Discussion

We recruited 20 participants (6 females and 14 males) with a mean age of 23.1 years (SD: 6.28). All of them were right-handed. All participants did not join the previous study. Figure 7 presents the



Figure 6: (ab) The process of deploying sensors for individual participants, along with four examples of this deployment. (c) Participants sat at a desk with their hands resting on the armrests or in their laps, wearing the prototype and performing gestures as instructed by the laptop.

data on average completion time, error rate, and the frequency with which participants preferred each of the ten gestures studied.

5.2.1 Completion Time. In analyzing completion times by finger group, results showed that single-finger gestures like the index finger completed in 0.33 seconds (SD: 0.13) and the middle finger in 0.32 seconds (SD: 0.06), while two-finger gestures such as the index-middle and middle-ring took slightly longer at 0.34 seconds (SD: 0.07) and 0.32 seconds (SD: 0.07). Three-finger gestures like the index-middle-ring were completed in 0.38 seconds (SD: 0.21), with four-finger gestures averaging 0.38 seconds (SD: 0.09) to complete.

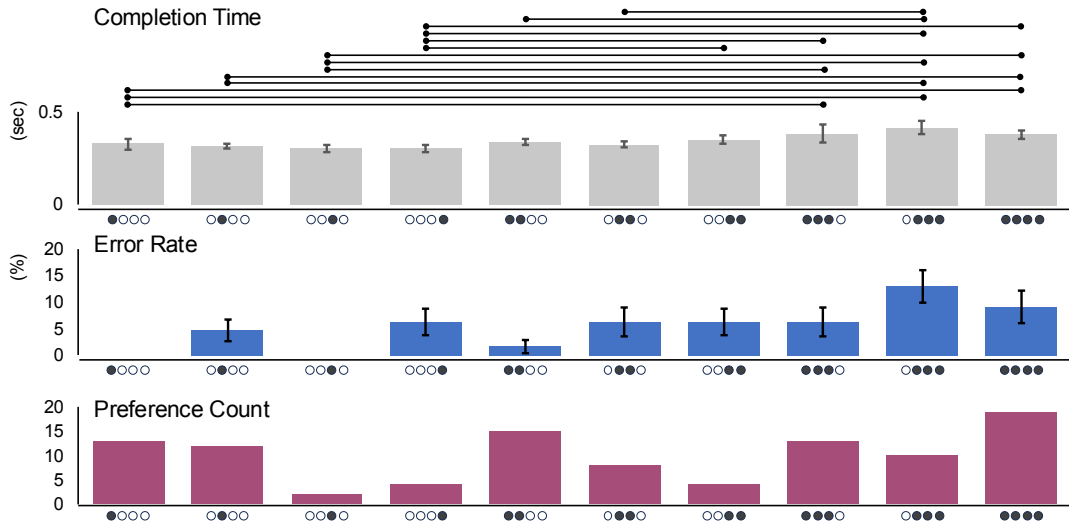


Figure 7: Results of the three methods on completion time, error rate, and preference count, with error bars representing 1 standard error. Pairwise significant differences are marked.

Using the Friedman test, a significant effect of gesture type on completion time was identified. Further analysis with the Wilcoxon signed-rank test confirmed that single-finger gestures were completed faster than those requiring three or four fingers, with a significance level of $p < 0.001$. No significant differences were found in completion times within groups of the same number of fingers, indicating a direct relationship between gesture complexity and time required.

5.2.2 Error Rate. The analysis of error rates segmented by finger group disclosed the following findings: In the single-finger category, the observed error rates were: the index finger gesture led to an error rate of 0% (SD: 0), the middle finger gesture 4.76% (SD: 0.09), the ring finger gesture 0% (SD: 0), and the pinky finger gesture 6.25% (SD: 0.11). For the two-finger category, the average error rates recorded were 1.63% (SD: 0.05) for the index-middle finger gesture, 6.25% (SD: 0.12) for the middle-ring finger gesture, and 6.25% (SD: 0.11) for the ring-pinky finger gesture. In the three-finger group, the error rate was 6.25% (SD: 0.12) for the index-middle-ring gesture and 13.04% (SD: 0.13) for the middle-ring-pinky gesture. Finally, the gesture involving all four fingers presented an error rate of 9.09% (SD: 0.13).

The error analysis across different gesture types, conducted using Friedman test, indicated a significant main effect. However, subsequent pairwise comparisons revealed no discernible differences among the gesture types. To delve deeper into the nature of these errors, we first examined the discrepancy between the errors and the correct answers. This examination revealed that all errors deviated by a single finger from the correct response. Further analysis of these errors suggested two primary underlying causes.

Finger Interdependence. The study of individual finger movements reveals the significant phenomenon of finger interdependence, particularly among the middle, ring, and pinky fingers, with these fingers often moving inadvertently when attempting to bend

one [12]. In contrast, the index finger shows greater independence. This interdependence can lead to inaccuracies in gesture recognition by causing unintended additional contacts. Common recognition errors include middle finger only gestures being incorrectly recognized as involving both the middle and ring fingers due to unintended ring finger movement, ring and pinky finger gestures misinterpreted as including the middle finger, and pinky finger only gestures erroneously recognized as involving both the pinky and ring fingers due to their interdependence. Additionally, index, middle, and ring finger gestures can be incorrectly identified as full hand gestures due to an unintended extra touch by the pinky finger, ring finger only gestures might be mistakenly identified as involving both the middle and ring fingers due to accidental middle finger touch, and index and middle finger gestures could be wrongly recognized as including the ring finger.

Finger Asynchronicity. When executing gestures, users often place their fingers on the palm at different times, which can lead to recognition errors if the timing varies by more than a predetermined 500-millisecond threshold in our algorithm. These errors are primarily due to unsynchronized finger movements. Although practice can improve synchronization, common discrepancies still occur. For example, middle-ring-pinky gestures are often wrongly recognized as just ring once, middle-ring twice, and Ring-pinky four times. Full gestures are frequently misidentified as index-middle-ring four times and middle-ring-pinky twice. Index-middle gestures may be incorrectly recognized as just middle once. Ring-pinky gestures are sometimes mistaken for just ring twice and pinky once, and index-middle-ring gestures are erroneously identified as just index once.

5.3 Preference

The study showed a clear preference for certain hand gestures, with the full-finger gesture emerging as the most favored, receiving 19

votes. It was followed by the index+middle gesture with 15 votes, and the index alone and index+middle+ring gestures each receiving 13 votes. Gestures involving the middle finger combinations garnered fewer votes, in descending order. Conversely, gestures utilizing fewer fingers, such as the pinky alone and ring-pinky combination, received the least preference, with only 4 and 2 votes respectively. Participants appreciated the full-finger gesture for its ease and likened it to an "all-in" action, describing it as natural and simple.

There was also a preference for gestures using just the index and/or middle fingers, attributed to their frequent use in daily tasks and common multi-touch gestures. However, gestures requiring three or four fingers were described as "effortless" due to the physiological advantage of distributing movement across the hand's musculature, reducing muscle fatigue. Less favored were gestures involving the pinky and ring fingers due to their interdependency and the challenges in isolating these fingers, which participants found unfamiliar and awkward, thus impacting gesture accuracy. The ring-pinky combination was particularly challenging, highlighting the difficulties with using less dominant fingers.

6 Design Guideline

Based on our results, we suggest the following guidelines to improve gesture-based interaction in applications: (1) Avoid Non-sequential Finger Movements: Study 1 highlighted the need to avoid non-sequential finger movements in gesture design due to the significant disparity in perceived physical and mental effort and lower confidence compared to sequential movements, stemming from anatomical challenges. (2) Prefer Sequential Finger Use: Gestures using three or four fingers were seen as requiring less physical and mental effort and had higher confidence levels among participants. Although these gestures involve longer execution times and are prone to errors from finger asynchrony, they are preferred, with potential errors manageable through algorithmic solutions. (3) Be Cautious with Ring and Pinky Fingers: Using the ring and pinky fingers independently is physically demanding due to strain on adjacent fingers and involuntary movements. Anatomical and neurological studies suggest these fingers naturally move in sync, complicating independent control. (4) Recommended Gesture Set: Based on our studies, we recommend seven gestures: full-finger, index-middle-ring, middle-ring-pinky, index-middle, middle-ring, index finger, and middle finger. While ring-pinky, ring alone, and pinky alone gestures are not excluded, they should be used less frequently in applications to avoid discomfort or injury.

7 Limitations and Future Work

The execution of finger-to-palm gestures may vary among individuals, particularly for those with thicker or shorter fingers, posing challenges in achieving secure contact. In our implementation, a uniform threshold was applied for detecting finger touches. However, employing finger-specific thresholds learned through machine learning could improve touch detection sensitivity. In addition, future research could consider shape-forming tactile interfaces, such as pneumatic skin interfaces [29], to improve this. Additionally, investigating the long-term effects on hand fatigue and muscle health

is vital. Currently limited to touch, expanding finger-to-palm gestures to include motion could enhance input options, though hand ergonomics must be carefully considered to avoid fatigue from repetitive movements. It is crucial to balance utility and comfort in gesture design, potentially through elicitation studies to identify natural gesture mappings for specific HMD functions, ensuring intuitive use and minimizing fatigue risk.

8 Conclusion

We introduced the "Finger-to-Palm interface," a new touch interface on the palm for improving interaction in HMDs using finger-to-palm commands. This method allows single-handed activation of HMD functions without needing precise targeting. Through two studies, we identified and recommended seven optimal gestures for this interface. The Finger-to-Palm interaction represents a novel single-handed control approach that could simplify user interactions in the advancing field of head-mounted computing.

Acknowledgments

This research was supported by the National Science and Technology Council of Taiwan under grants 113-2223-E-A49-006-MY3 and 113-2628-E-A49-013-MY2.

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